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A model for the Holocene extinction of the mammal megafauna in Ecuador

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Abstract

This paper presents the results of multidisciplinary research in the Ecuadorian coastal regions, with particular emphasis on the Santa Elena Peninsula. The new evidence, together with previous data gathered on the Ecuadorian cordillera during the last 12 years, allows us to formulate a model that accounts for most of the mammal megafauna extinction at the Pleistocene/Holocene transition. After the illustration of geomorphological and paleontological evidences of the area of the Santa Elena Peninsula (and other sites), and of a summary of the paleoclimatic data, the main results and conclusions of this work are: (1) Late Pleistocene mammal assemblages survived in the Ecuadorian coast until the Early Holocene sea level rise; (2) Prior to the extinction of most of the megafauna elements (mastodons, ground sloths, equids, sabre-tooth felids), the mammal communities at Santa Elena Peninsula comprise elements with differing habitat requirements, attesting conditions of high biological pressure; (3) At the El Cautivo site (Santa Elena Peninsula), we have discovered Holocene sediments containing the first known occurrences in Ecuador of lithic artifacts that are associated with mammal megafauna remains; (4) During the last 10,000 years, the coastal region of Ecuador underwent significant changes in vegetation cover. At the Pleistocene/Holocene transition the climate changed from very arid conditions to humid conditions.

Our data indicates that the megafauna definitively abandoned the Cordillera areas around 12,000 yr BP due to t he increasing aridity, and subsequently migrated to coastal areas where ecological conditions still were suitable, Santa Elena Peninsula and mainly Amazonian areas being typical. We conclude that the unusual high faunal concentrations and the change to dense vegetation cover (due to a rapid increase in precipitation in the lower Holocene) at 8000–6000 yr BP, caused the final collapse and extinction of most elements of the mammal megafauna. Vegetation cover in the area of Santa Elena should have been extensive, and even more so in the Guayas and Guayabamba valleys. The newly densely vegetated areas, and fluvial barriers, transformed the refugia into lethal traps for large animals already under biological stress, such as were mastodons, ground sloths and equids. Within the megafauna, only tapirs and artiodactyla (Cervidae and Camelidae) survived.

In our opinion, the most suitable model to justify the great crisis of the mammal megafauna at the Pleistocene/Holocene transition, also in areas out of Ecuador, must be mainly based on the three parameters: high aridity, high humidity and geographic factors.

Keywords: Mammal; Megafauna; Extinctions; Latest Pleistocene; Earliest Holocene; South America

Resumen

En este artículo se presentan los resultados de una investigación multidisciplinaria en las regiones costeras de Ecuador (especialmente en la Peninsula de Santa Elena). Nuevas evidencias, junto con los datos de doce años de investigación en la Cordillera Andina, permite formular un modelo explicativo sobre la extinción de la megafauna de mamíferos en la transición Pleistoceno–Holoceno. A partir de las evidencias geomorfológicas y paleontológicas del área de la Península de Santa Elena (y de otros lugares), y de los datos paleoclimáticos, se concluye que: (1) las asociaciones de mamíferos del Pleistoceno tardío sobreviven en la costa ecuadoriana hasta que el nivel del mar sube en el Holoceno inferior; (2) previamente a la extinción de los elementos de la megafauna (mastodontes, perezosos, équidos, tigres de dientes de sable), las comunidades de mamíferos de la Penínula de Santa Elena comprenden elementos con diferentes necesidades de hábitat, que

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confirman condiciones de alta presión biológica; (3) en el yacimiento de El Cautivo (Península de Santa Elena), se han descubierto sedimentos que contienen las primeras evidencias conocidas en Ecuador de industrias líticas asociadas a restos de megafauna de mamíferos; (4) durante los últimos 10,000 años, se registran importantes cambios de la cobertura vegetal en la región costera de Ecuador. En la transición Pleistoceno/Holoceno el clima cambió desde condiciones muy áridas a condiciones húmedas.

Se considera que la megafauna abandonó definitivamente las áreas de la Cordillera en torno a 12,000 años BP, y subsecuentemente emigró a las áreas costeras donde se formaron nuevas asociaciones faunísticas. Posteriormente, conforme la cobertura vegetal se incrementó, la megafauna se fue concentrando en áreas aisladas donde las condiciones ecológicas todavía eran adecuadas. La Península de Santa Elena y las principales zonas amazónicas son las más típicas. Aquí sin embargo, las inusuales altas concentraciones de fauna y el cambio hacia una densa cobertura vegetal (por el rápido incremento de las precipitaciones) entre 8000 y 6000 años BP, causaron el colapso final y la extinción de la mayoría de los elementos de la megafauna de mamíferos. La cobertura vegetal en el área de Santa Elena debe haber sido densa, incluso más en los valles de Guayas y Guayabambas. Las densas áreas forestadas nuevas y las barreras fluviales, transformaron los refugios en trampas letales para los grandes animales todavía bajo condiciones de estrés, como sucedía con mastodontes, perezosos y équidos.

Se considera que el modelo explicativo para la extinción de los elementos de la megafauna en la la transición Pleistoceno/Holoceno tiene que ser articulado sobre tres parametros: alta húmedad, alta áridez y el factor geografico.

Palabras clave: Mamíferos; Megafauna; Extinción; Pleistoceno tardío; América del Sur

1. Introduction

Recent studies on late Pleistocene mammal associations from the Ecuadorian Cordillera have pointed out a sequence of megafauna migratory events. During the last glacial maximum the mastodons were the first megafauna to disappear as a consequence of climatic deterioration. Later, in a relatively shorter interval, the same happened for the xenartrs (*Glossotherium*) and the equids (Ficcarelli et al., 1997). These different dispersal events were dated also radiometrically (^{14}C) thus allowing the creation of a temporal model for the megamammals' extinction at the Pleistocene/Holocene transition (Coltorti et al., 1998). It was hypothesized that increasing aridity and cooling during the Late Pleistocene forced these animals to migrate from the Cordillera in search of more suitable environment. One megamammal refugium was hypothesized in the Santa Elena Peninsula (Fig. 1) where a rich faunal association was collected and attributed to the Upper Pleistocene (Hoffstetter, 1948, 1952; Sauer, 1965). In our model the extinction of many of the megafauna species found in the refugium is hypothesized as due to the dramatic biological stresses arising from anomalous faunal concentrations. We also recognized that in such an unstable situation, locally, a secondary critical role could have been played also by human activity.

To test this model, we attempt to improve the dating of the deposits on the Santa Elena Peninsula, find proof of human presence associated with the mammal remains or in the stratigraphically correlated sediments, identify and date any other coastal fossiliferous sites, and identify the vegetation co-occurring with the megamammals in order to describe the climatic setting during the time of catastrophic crisis. Therefore, a multidisciplinary investigation has been carried out from 1998 to 1999 on the Santa Elena Peninsula (Guayas Province) and in the adjacent coastal regions of Ecuador. We present here the new data for a better understanding of the stratigraphic setting and the geochronology of the continental megamammal associations of Ecuador during this period of rapid change of the ecosystems. Finally, we propose a new more exhaustive model to explain the megamammal crisis at the Pleistocene/Holocene transition.

Previous authors have identified significant paleontological sites from La Carolina (today called La Libertad) in the Santa Elena Peninsula, but their stratigraphical settings have not been firmly established (Spillmann, 1942; Hoffstetter 1948, 1952). Spillmann (1942) recognized three successive fossiliferous levels, the lowermost containing large animals and the uppermost containing the smallest ones, but was uncertain of the age of the deposits. In contrast, Hoffstetter (1948, 1952) attributed all findings of the La Carolina faunal assemblage to a single stratigraphical level of Upper Pleistocene age and proposed a similar Upper Pleistocene chronostratigraphical setting for other deposits of the coastal region, such as Rio Daule Basin, and Puna island. Almost all the faunal remains are preserved in the Museum of the Departamiento de Biologia of the Escuela Politecnica Nacional of Quito.

During our recent surveys, we have studied in more detail the El Cautivo site in the Santa Elena Peninsula, which was initially discovered and studied by researchers from the Escuela Politecnica del Litoral of Guayaquil (Wunch and Piquè, 1995). We also discovered new fossiliferous sites south of Machalilla and to the south of Manta in San Jose (Cantalamessa et al., 2001). Our new investigations improve the chronological setting of the vertebrate faunal assemblages from the coastal area, thereby enabling a more accurate correlation of the dynamic relationships with the assemblages from the Cordillera.

2. Geological and geomorphological setting

The Ecuadorian coast is characterized by low relief with a mean altitude of about 600 m a.s.l.; maximum elevations

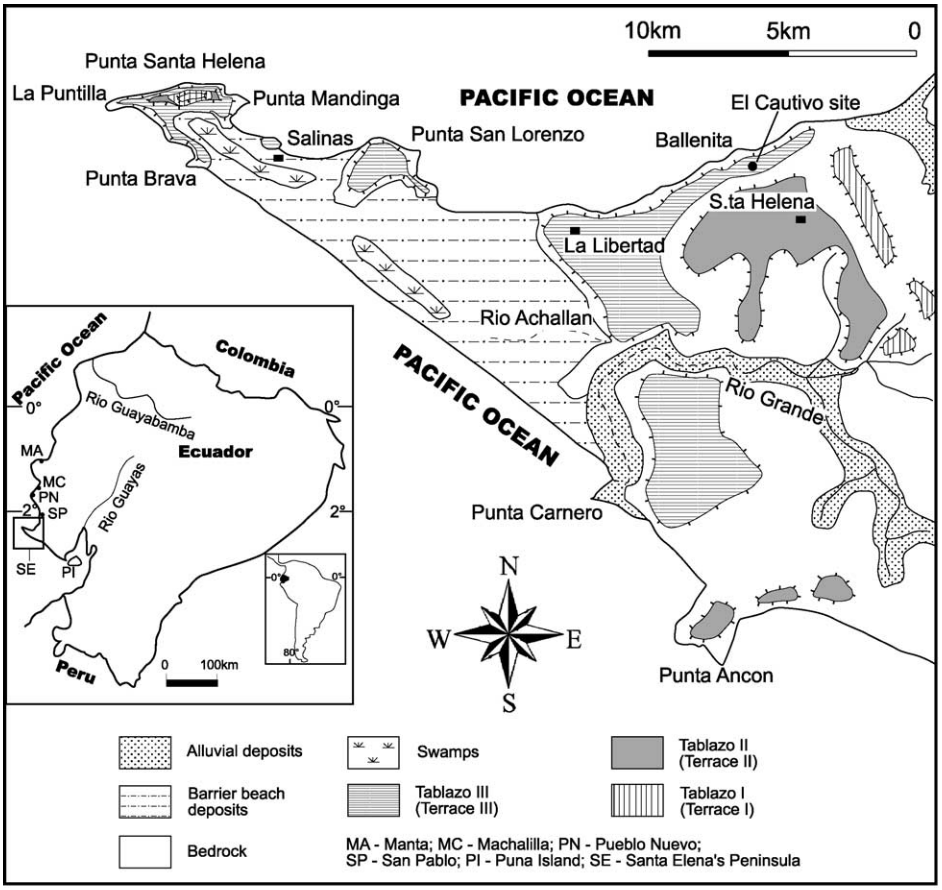

Fig. 1. Geomorphological map of the Peninsula of Santa Elena (Ecuador).

of 1000 m a.s.l. are found in the Jama Hills to the north and in the Colonche-Chongon Hills to the south (Baldock, 1982; Litherland et al., 1993). East of these hills is a relatively lower topographical depression into which rivers that originate in the Andean highlands drain. The main river is the Rio Guayas, which flows southward into the Guayas gulf and creates one of the largest estuaries on the Pacific side of South America (Ayon and Jara, 1985).

Present-day climate and ecosystems are strongly influenced by the presence of the Intertropical Convergence Zone. The Ecuadorian coast is close to the convergence between the Humboldt (or Peru) cold current and the warming southern equatorial current. The seasonal extremes of mean sea surface temperature (SST) can vary from 2 to 10 °C, with an irregular periodicity of 2–8 years, in connection with the El Niño-Southern Oscillation (ENSO) phenomenon. The SST variations are associated with heavy but regionally discontinuous precipitation events, even at high elevations (Rodbell et al., 1999).

2.1. The Santa Elena peninsula

The Santa Elena Peninsula is located on the southwestern coast of Ecuador and is characterized by less rain when compared to the rest of the coast. Present drainage consists of streams that are dry for most of the year. The main watercourse is the Rio Grande, which enters the Pacific Ocean close to Punta Carnero. Hoffstetter (1948) suggested that the Rio Grande might have flowed northward into the Rio Achallan since a saddle is located to the west of the low relief of La Represa. Even up to the present, the fast growth of barrier beach deposits on the peninsula's western side has periodically dammed the river's mouth thus forcing its diversion to the north.

The present-day coastal morphology of the peninsula is asymmetric. The northern coast is characterized by headlands separated by cove beaches, whereas on the southwestern coast the beaches are more continuous and interrupted by relatively smaller headlands. Extensive lagoons, which have become more saline in succession, characterize the back-shore deposits that are found west of La Carolina and along the entire south western exposure. The lagoon was dammed in the past by barrier beaches growing initially between La Carolina and Santa Rosa to the north, and Punta Brava and Punta Carnero to the west. In the Early Holocene, the Santa Elena Peninsula likely extended no further west than present-day Punta Carnero; however, emerged land already existed between La Puntilla and Santa Rosa. The successive development of a spit connected these islands to the mainland.

The present-day coastal dynamics are strongly asymmetric. The greatest sediment accumulation is found on the south western reach due to littoral currents that rework and transport sediments brought in by the Rio Grande. The Guayas River sediments may have been similarly transported along the shore.

Inland, the morphology has been strongly influenced by repeated regressive–transgressive cycles, which take place in a context of overall tectonic uplift. Three extensive marine terraces (Tablazos) are visible almost everywhere on the peninsula. Their marine origin was first postulated by Hoffstetter (1948). Marchant (1961) suggests that the three terraces were originally a single terrace that was later displaced by faulting, however, our study supports Hoffstetter's (1948) conclusion.

During marine transgressive phases, the marine abrasion platforms were formed, on which thin coastal deposits were deposited. During the regressive phases, these platforms were incised by streams creating a network of gullies that flowed toward the Pacific Ocean, which at that time was a few tens of kilometers to the northwest. The oldest and highest terrace, exposed to the east of Santa Elena and at La Puntilla (Tablazo I), is found at 75–80 m a.s.l. These sediments are isolated and strongly dissected by erosive processes. The intermediate terrace (Tablazo II), is located at an elevation of about 35–40 m a.s.l. and is the most evident terrace of the three. The town of Santa Elena is situated on it. The lowest terrace (Tablazo III) is usually located at about 10–15 m a.s.l. though it sometimes outcrops just above the present-day sea level. It is recognizable on the peninsula's northwestern coast and, in a thin strip close to the southwestern coastline. However, close to some of the headlands, behind the present-day coastline, an active platform of marine abrasion is also present. This platform is located just above the mean present-day sea level and landward, it ends in a notch at the foot of the headlands. At Punta Carnero however, two superimposed notches appear. The overall uplift of the area, in part documented by these terraces, and the Holocene environmental changes have been previously recognized by Estrada (1961), Stothert (1983), and Damp et al. (1990).

The mammal fauna of the La Carolina and El Cautivo sites have been found within the lithostratigraphic unit that characterizes the uppermost terrace (Tablazo III) that Hoffstetter (1948, 1952)) attributed to the Late Pleistocene.

A detailed study of these sediments (Fig. 2) shows great variability of the sedimentary facies. High-energy coarse sandy beach deposits characterize the southwestern coast between Punta Cranero and La Puntilla, the northern shore between Punta Mandinga and Punta San Lorenzo, and east of La Ballenita up to San Pablo. In these localities the sandy beach deposits are strongly cemented and usually outcrop at the same elevation of the present-day foreshore. They are therefore being actively eroded by present-day dynamics. Elsewhere (most evident between La Ballenita and Punta Mandinga), this sedimentary unit is represented by estuarine deposits, made of alluvial to coastal lagoon facies that pass seaward into barrier-beach deposits. The alluvial deposits,

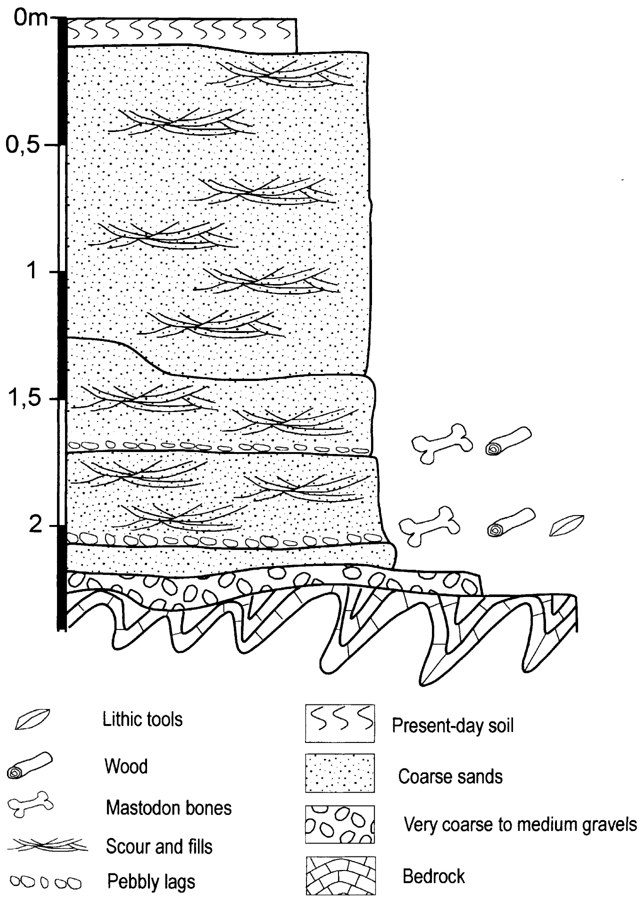

Fig. 2. Stratigraphical log of the site El Cautivo (Peninsula of Santa Elena, Ecuador).

composed of coarser grained facies, are most evident in correspondence to the mouths of the old watercourses. The coastal lagoon facies are characterized by finer grained sediments that frequently contain calcified rhizomorphs that likely originated along mangrove roots, as suggested for areas to the north (Norton et al., 1983). Marine molluscs, indicative of an estuarine or lagoon environment with mangrove vegetation (i.e. *Anadara tubercolosa*, *Ostrea* sp., *Chlamys* sp., *Cerithidea* sp.) are ubiquitous within these sediments (Hoffstetter, 1948).

This suggests that the Early Holocene transgression in part flooded the valleys that had been incised in the Pleistocene. The rising sea level generated small protected coves or swamps with mangrove vegetation that were fed both by ground and running water. The top of this sedimentary unit (Tablazo III) can be found at elevations of up to 18 m a.s.l. However, the barrier beach systems developed during the latest phases of deposition. A similar pattern of Early Holocene marine sedimentation has been identified in Perù (De Vries and Wells, 1990), in which the initial formation of small estuaries later took the form of low beach ridges that prograded across a transgressive marine platform. All along the Santa Elena Peninsula, the growth of barrier beaches generated lagoons of varying dimension;

the largest beach was delimited by a triangular tombolos located between La Carolina, Punta Carnero and La Puntilla.

The presence of the molluscs, which are typical of mangrove swamps and always found associated with the megamammal fossils, strongly implies that the faunal assemblage belongs to the Holocene. The molluscs at the El Cautivo site have been dated AMS ^{14}C to 8680 ± 80 yr BP (sample ETH-20255; calibrated, age corresponding to 7914–7541 yr BC). However, since the bones lack collagen additional radiocarbon analyses were not possible. The associated sediments were barren of pollen, and so correlation based on vegetation assemblages was also not yet possible.

In the levels with megafauna bones, only a few meters distant, the presence of lithic tools (Fig. 3) indicate humans. Wunch and Piquè (1995) note lithic artifacts associated with the megafauna remains for the El Cautivo site. The Las Vegas culture, which according to Stothert (1985) is approximately 9000–7000 calendar yr BP in age, and whose type locality is located a few kilometers to the south, should be contemporaneous with the industries found at El Cautivo. Unfortunately, since only a few tools have yet been identified at El Cautivo, so an accurate comparison with the Las Vegas lithic assemblages is not possible. However, the El Cautivo site contains the oldest artifacts and tools associated with the megafauna of Ecuador.

The people of the Las Vegas culture exploited the ecosystem just described; molluscs particular to the mangrove swamps (*Anadaria tubercolosa*) were approximately one-quarter of their diet (Stothert, 1985). Molluscs of about the same age and associated with the mangrove swamps, have been recognized in north central Peru (e.g. the Talara, and the Amotape Cultures assigned to 11,000–8000 calendar yr BP; Sandweiss et al., 1996). However, megafauna remains have never been found in either the Las Vegas or in the southern Peruvian archaeological sites.

After the formation of Tablazo III during the early Holocene, coastal and inland dynamics underwent major changes in response to multiple forcing. The existence of terraces up to 18 m a.s.l. indicates that tectonic uplift has affected, and likely is still affecting, the entire peninsula. Most of the stream valleys are hanging because of this uplift. Only a few watercourses had vertical incision rates that could balance the uplift rates that originated coastal plains of limited extent, but they are usually dry and filled with sediments. Damp et al. (1990) have suggested that uplift rates varied significantly; in particular they recognize an abrupt uplift event at approximately 3700 ^{14}C yr BP. Similar events may have caused important changes throughout the Peninsula.

The entire northern coast of the Santa Elena Peninsula, from Punta Ballenita to La Puntilla, is being severely eroded. A marine abrasion platform, extending seaward up to several hundred meters, is now being formed. On the headlands, usually close to the sea level, but locally also occurring few meters above sea level, deep erosional notches are now being formed within the Punta Carnero rocks.

In general, coastal morphology and the transition from an estuarine coasts to a barrier beach system depends on the interaction among uplift processes, eustatic movements, and sedimentary processes and their changes in time (Valentin, 1952; Pirazzoli, 1991, 1996). In the Santa Elena Peninsula, the uplift processes are ongoing, as is evidenced by the elevation of the Holocene deposits and the data presented by Damp et al. (1990). However, the greatest sea level rise apparently took place earlier than that observed for the northern hemisphere (Pirazzoli, 1991). Although sea levels are not thought to have stabilized globally until 6000–5000 yr BP, our data indicate that an important transgressive phase started just before 8000 ^{14}C yr BP. In general, the combination of tectonic uplift and positive eustatic movements would work against the creation of beaches and lagoons. Nevertheless, the sedimentary processes may have played an important role on the Santa Elena Peninsula. Possibly during the Middle Holocene, the sediment load of the watercourses was large enough to override the eustatic and tectonic dynamics, which resulted in the growth of barrier beaches.

Moreover, in spite of the recent and ongoing rapid uplift, beaches are still prograding in some sectors of the peninsula.

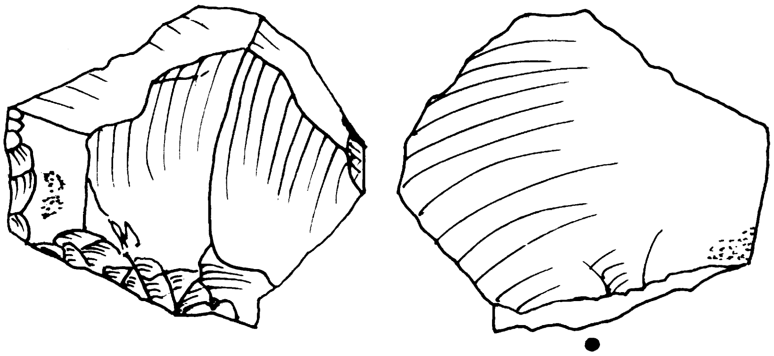

Fig. 3. A lithic tool associated with paleontological remains from El Cautivo (Peninsula of Santa Elena, Ecuador).

The causes of the rapid sedimentary processes can be several. Deposition related to fluvial transport may have been important after sea levels stabilized in the mid-Holocene. However, sediment supply also may have increased due to soil erosional processes related to agricultural and pastoral deforestation. The people of the Las Vegas Culture probably did not induced important environmental changes, because they survived by hunting and fishing. The Neolithic populations of the Valdivia Culture (5500 calendar yr BP), among the oldest of South America (Evans et al., 1959), and successive populations (Marcos, 1986), practised agriculture, however. Moreover, deforestation has accelerated since the 19th century in association with petroleum excavation and the ensuing increases in population (Meggers, 1966).

Owing to these recent processes related to anthropic activities, the remaining coastal lagoons have been transformed into saltpans or artificially filled and reclaimed. A few lagoons have been dammed and used as reservoirs. Without the changes induced by human activities a tropical forest would cover the area and ensure protection against soil erosion. Modern agricultural practices and the recent abandonment of animal husbandry is now permitting a progressive recolonization of the vegetation. The surroundings of Santa Elena are currently characterized by a closed-canopied forest.

2.2. Pueblo Nuevo (Machalilla) and San Jose (Manta) sites

The fossiliferous site of Pueblo Nuevo is located a few hundred of meters from the present-day coastline, to the north of Machalilla (Fig. 1). Fossil bones were found in the middle of terraced alluvial deposits cut by an active small stream that flows directly into the Pacific Ocean. The sediments are composed of subrounded to subangular gravels (Fig. 4). Troughs cross-bedding with a few planar cross-bedding features within shallow channels of limited lateral extent are indicative of braided fluvial deposits (Miall, 1985, 1996). A few longitudinal bars were also observed. In the middle portion of these deposits, sandy and silty layers evidence an abrupt decrease of bedload. Close to the top of the deposit, at approximately 15 m above the thalweg, a thick soil is truncated and buried under layers related to occupation by the Valdivia culture.

The present-day watercourse is diverted to the south around a hill, just before reaching the ocean, creating an epigenetic valley. This valley and its infilling hang on the nearby coastal cliff. Ongoing rapid coastal uplift (similar to that at Santa Helena) creates a marine erosional platform approximately 1 m a.s.l. Despite the close proximity of the fluvial deposits to the coastline, the upper surface of the depositional terrace exhibits a very low angle slope of ca. 2°. The slope surface's preservation up to 15 m above the cliff suggests strong coastal retreat after deposition of the alluvium and after the Early Holocene sea level rise. The braided alluvial deposits indicated that the slopes within the drainage basin were bare, without vegetation and affected by intense soil erosion processes during arid periods of the Upper Pleistocene (Colinvaux and Scofield, 1976; Clapperton, 1993). These processes ceased at the beginning of the Holocene, when the present-day tropical forest colonized the area.

The reforestation and sea-level rise are responsible for the deepening of the valley, the terracing of the deposits, and the development of soils that buried the terraces. Archaeological findings from the Early Holocene at the top of the terrace confirm an older age for these terraced deposits.

During the most recent investigations of the coastal region, a new vertebrate fossil site was found at San José (Cantalamessa et al., 2001). The deposit consists of estuarine sediments containing marine mollusc fauna; it fills a valley that is now uplifted, and crops out along a cliff, 5–10 m above the present-day beach. It constitutes a morpho- and lithostratigraphical unit that might correspond to the Tablazo III unit observed on the Santa Elena Peninsula. These considerations allow a probable Early Holocene age attribution to the faunal assemblage (Cantalamessa et al., 2001).

3. Palaeontological evidence

The geopaleontological field survey of the Santa Elena Peninsula allow us to understand the significance of the El Cautivo site (Wunch and Piquè, 1995). The deposit clearly correlates with those first recognized in the La Carolina site, upon which the city of La Libertad is built. The new site has yielded remains of *Haplomastodon chimborazi*, *Equus santa elenae* and *Glossotherium* sp. As already stated by Hoffstetter (1952), the vertebrate fossil fauna of the Santa Elena Peninsula is apparently concentrated in a single fossiliferous level. Together, the older collections and the newer findings, depict a vertebrate fauna represented by rare amphibians, squamates, abundant chelonians (mainly large tortoises), aves and several mammals, such as *Haplomastodon chimborazi*, *Eremotherium laurillardi* or *E. rusconii*, *Scelidotherium reyesi*, *Glossotherium tropicorum*, *Chlamytherium occidentale*, *Equus santa elenae*, *Palaeolama aequatorialis*, *Odocoileus salinae*, *Neochoerus sirasakae*, *Echimyidae*, *Dusicyon sechurae*, *Protocyon orcesi*, *Puma* sp., *Smilodon* sp.

The association is very significant because it contains elements of forest origin, such as mastodons, megatherians and deer, as well as elements typical of open habitats, such as equids and camelids. We believe taphonomic bias (in which horizontal reworking leads to a mix of animals from different habitats) is an unlikely explanation for the association, because the material is well preserved and the watercourses are short. More likely, this association is linked to a relatively unusual environment that persisted for some time in the

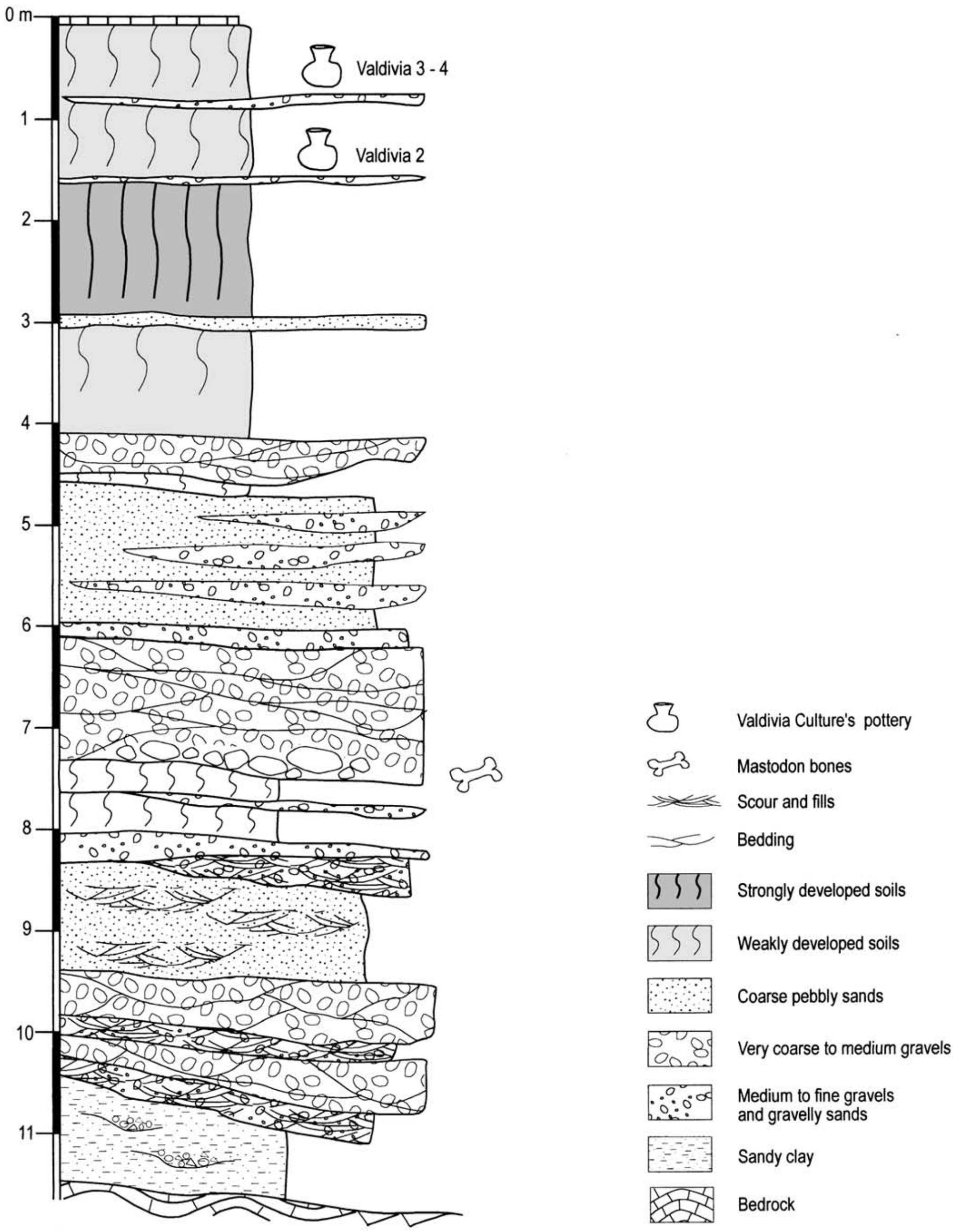

Fig. 4. Stratigraphical log of the site Pueblo Nuevo (Machalilla, Ecuador).

coastal areas of the Guayas province, and it may be evidence of a refugium during the climatic deterioration of the last glacial maximum.

The composition of the faunal assemblage might be attributed to the Late Pleistocene instead of the earliest Holocene, but the AMS-^{14}C age of *Anadara tubercolosa* shells (8680 ± 80 yr BP) confirms the latter.

A vertebrate fauna represented by *Haplomastodon chimborazi*, *Eremotherium laurillardi* (or *E. rusconii*) and *Geochelone* sp. was collected near Pueblo Nuevo, south of Machalilla, during our survey of the coast north of the Santa Elena Peninsula. The same considerations as used for the Santa Elena assemblages could apply to this locality, but in this case the geological settings suggest a Late Pleistocene age.

Farther north, in San Jose, we identify vertebrate fauna similar to that described for Pueblo Nuevo. On the basis of the geological setting, it must be attributed to the Early Holocene (Cantalamessa et al., 2001).

The vertebrate faunal assemblages of the Ecuadorian coastal area, with the exception of those from the Santa Elena Peninsula, are poorly represented. The lack of findings is a great obstacle when reconstructing the fauna's evolution during the Upper Pleistocene–Early Holocene time span, and when comparing the coeval mammal communities of the Ecuadorian Cordillera.

The primary problem is systematic. It is difficult to evaluate the importance of certain morphological differences in related taxa from the two areas. The differences may express a specific systematic status, or they may simply reflect ecophenotipic morphologies. Detailed systematic analysis of the mastodons (Ficcarelli et al., 1993, 1995) and deer (Abbazzi and Tomiati, in press) suggest the second hypothesis is more likely but further analyses of the fauna and more reliable dating of some coastal sites are required.

Despite the uncertainties, some interesting results should be emphasized, including the absence of *Eremotherium* (the existing published data contain misidentifications), *Scelidotherium*, *Chlamytherium* and *Neochoerus* in the deposits of the Ecuadorian Cordillera, and the presence of *Haplomastodon chimborazi* both in the Late Pleistocene Cordillera sites and in Late Pleistocene–Early Holocene coastal sites. This last detail indicates the widespread range of the mastodons, their successful cohabitation in different biological communities and hence their great ecological flexibility.

Two different scenarios are proposed to account for the faunal interactions between the Cordilleran and the coastal assemblages. Both scenarios could lead to identify the last Holocene refugia of several of the mammal megafauna, in territories with ecological characteristics similar to those found on the Santa Elena Peninsula.

If the morphological differences between the Cordillera and Costa faunas reach the systematic level, then we need to propose different reason and times for the faunal extinction in the two areas. The Cordilleran faunal crisis would be due to progressive shrinking of the species' preferred ranges due to increasing aridity and climatic cooling during the Late Pleistocene, which would lead to the extinction of some megafauna elements (Ficcarelli et al., 1997; Coltorti et al., 1998). In coastal areas, such as the Santa Elena Peninsula, despite the biological pressure, different megafaunal elements survived until the Early Holocene. At that time abnormal increasing humidity and the ensuing forest overgrowth made the environment a trap.

If however, the faunal differences between the Cordilleran and the coastal populations were only ecophenotipic adaptations, then the faunal crisis leading to the Holocene extinctions is more likely a long and continuous process with a sequence of faunal dispersal events from the Cordillera toward the Coast connected to late Pleistocene climatic deterioration. In the coastal areas, the same environmental conditions became fatal.

We favor the second hypothesis which is more consistent with the faunal fossil evidence found in different areas of the country, and more strictly accounts for the biological requirements. Therefore, this second hypothesis represents the base for the explanatory model of the Early Holocene extinction of mammal megafauna discussed subsequently.

4. Paleoclimatic data

The available Ecuadorian paleoclimatic data come from lacustrine sediments of the Cordillera and the Galapagos areas (Colinvaux and Scofield, 1976; Colinvaux et al., 1988; Kuhry, 1988; Hansen and Rodbell, 1995; Hansen 1996), though further data are available in Perú from drills in the Huascaran glaciers (Thompson et al., 1995), and from ice-cores in Bolivia (Thompson et al., 1998). The common characteristic is a rapid shift from a very arid to a wet condition at approximately 10,000 ^{14}C yr BP, which affected the coastal regions of Peru and Ecuador (Colinvaux et al., 1988). In the Ecuadorian Cordillera, important Holocene climatic changes have not been recorded, though minor changes of the amount of precipitation have been observed historically. Drilling in the Huascaran glaciers, at approximately 9°S, revealed mean temperatures higher than those of today in the 8400–5200 yr BP interval, with a peak between 6200 and 5200 yr BP. Comparable data resulted from ice-cores in Bolivia. In the Colombian Cordillera, warm and wet Holocene conditions lasted up to 2000 yr BP, and were followed by drier (and cooler?) climate (Kuhry, 1988). In the Peru Cordillera, a shift toward more arid conditions is recorded later than 6000 yr BP, even if anthropogenically induced environmental changes start to be sensible after this date (Hansen and Rodbell, 1995). For example, the cultivation of *Zea mays* is found as early as 8000 yr BP in the Columbia Cordillera (Kuhry, 1988) whereas in the Amazon lowlands of Ecuador it is found at about 6000 yr BP (Bush et al., 1989).

The study of the molluscs faunas (Hoffstetter, 1948) in the archaeological sites of Ecuador (Sarma, 1974; Stothert, 1985) and Peru (Rollins et al., 1986; Sandweiss et al., 1983, 1996, 1999; De Vries and Wells, 1990; De Vries et al., 1997) reveals important environmental changes in the coastal area soon after the last occurrence of the megafauna, though the causes of these changes are under debate.

Sarma (1974) discussed the discontinuous distribution of settlements on the Santa Elena Peninsula and the greater use of species typical of coastal lagoons with mangroves (i.e. *Anadara tubercolosa*). He suggested an alternation of wet and arid conditions during the Middle–Late Holocene: the wet periods would correspond to the archaeological records (Las Vegas, 9000–6500 yr BC; Valdivia I, 2450–2650 yr BC; Valdivia 3 through 6, 1800–2100 yr BC; Engoroy, 280–850 yr BC; Guangala 1, 50–100 BC), but the arid phases in between would be unfavorable for human settlements. However, the absence of evidence is not the evidence of absence, and the lack of archaeological sites could be attributed to accidental (Damp et al., 1990) or cultural (Zeidler, 1986) causes. Furthermore, the variable amount of molluscs in the archaeological sites could be attributed to the seasonal or cultural changes of the population's diet, as suggested by Stothert (1985).

The recent disappearance of the mangrove cover is probably a consequence of the human impact that led to important changes in the natural environments of the peninsula. Mangrove forests appear both to the north (Manglar Alto) and to the south (Guayas estuary). The mangrove trees easily survive in hypersaline water, without fresh water input, for most of the year (Richardson, 1980) but strongly suffer the increase of bedload and the rapid sedimentation, which are usually linked to the soil erosion processes that follow deforestation in the inland areas.

Thermically anomalous molluscs associations (TAMA) also have been noted along the Peruvian coast. North of 10°S the 10,000–5000 yr BP interval is characterized by tropical assemblages, whereas to the south, temperate assemblages are predominant (Richardson, 1980; Rollins et al., 1986; Sandweiss et al., 1983, 1996, 1999; Keefer et al., 1998). Tropical species are found as far north as Puemape, Peru, (7°30′S; Salinas culture, 2350–2050 yr BP), whereas variations of temperate species are observed at up to 12°S. This indicates that ENSO did not exist before 5000 yr BP (Sandweiss et al., 1996, 2001). A detailed study of the environmental context of these mollusc assemblages revealed that, in southern Peru, TAMA and temperate assemblages are found together from ca. 10,500 to 5000 yr BP (De Vries and Wells, 1990; De Vries et al., 1997).

Warm species could live inside protected lagoons because of the presence of barrier beaches. They moved at the larval stage during occasional ENSO, as is suggested for the Pleistocene deposits in California, USA (Zinsmeister, 1974). The same explanation can be invoked for the Ecuadorian faunas associated with arid climatic conditions (Sarma, 1974), suggesting the importance of the local variations of the physical environment.

5. Conclusions

The results of our multidisciplinary research are summarized as follows:

1. Late Pleistocene mammal assemblages survived in the Ecuadorian coast until the Early Holocene sea level rise. The fossils appear in estuarine deposits containing typical mangrove mollusc assemblages (i.e. *Anadara tubercolosa*);
2. Prior to the extinction of most of the megafauna elements (mastodons, ground sloths, equids, sabre-toothed felids), the mammal communities at Santa Elena Peninsula comprise elements with differing habitat requirements. This attests to conditions of high biological pressure;
3. During the last 10,000 years, the coastal region of Ecuador underwent significant changes in vegetation cover. At the Pleistocene/Holocene transition the climate changed from very arid conditions, characterized by mainly areal erosional dynamics and valley aggradation, to humid conditions, characterized by linear erosion and river downcutting;
4. At the El Cautivo site (Santa Elena Peninsula), Holocene sediments contain the first known Ecuadorian occurrences of lithic artifacts associated with mammal megafauna remains; and
5. The environmental changes at the Pleistocene/Holocene transition are driven by climatic changes. During the Holocene, the environmental conditions are mostly driven by the interaction between tectonic uplift and erosional/depositional processes, in part, induced also by human activity.

We have attempted to integrate the results of 12 years of multidisciplinary research in the Ecuadorian Cordillera and coast regions, into a single theoretical model that accounts for most of the mammal megafauna extinction at the Pleistocene/Holocene transition.

We believe that human impact cannot be solely responsible for such a global crisis, although it might have been a minor local factor. Similarly, toxic compounds emitted by vegetation under strong pressure from feeding herbivores remain a possible local factor in the extinction (Guthrie, 1984). The global environmental changes at the Pleistocene/Holocene transition are driven mainly by climatic factors. Our studies on the Santa Elena Peninsula detail how the vegetational belts of 8000 yr BP differ significantly from those of the present day. This change in vegetation plays an important role in resolving some crucial questions in our previously proposed model (Ficcarelli et al., 1997; Coltorti et al., 1998).

It was clear that the disappearance of mastodons, ground sloths and equids from the Cordillera occurred following the climate deterioration of last glacial maximum and that anthropic influences in the megafauna refugia along the coastal area were of secondary importance, with respect to the unusual biological pressure caused by the high faunistic concentrations (Ficcarelli et al., 1997; Coltorti et al., 1998). However, the causes of the final magafaunal extinction at around 8000 yr BP remain unclear.

Our new data and those in recent literature enable us to propose that a rapid increase in precipitation in the Early Holocene, over a short time interval, would induce dense forest cover which, along with the main fluvial belt generated a geographical barrier. Vegetation cover in Santa Elena should have been extensive, and even more so in the Guayas and Guayabamba valleys, two large river basins coming from the Cordillera.

The new densely vegetated areas, and fluvial barriers, transformed the refugia into lethal traps for large animals, such as mastodons, ground sloths, and equids, already under biological stress. Only artiodactyla (deer and camelids) and perissodactyla (tapirs) among the mammal megafauna survived, but this can be explained by the grater nutritive potentiality of the former and the more protective habitat preferences of the latter.

Similar phenomena, but at a larger scale, likely occurred in the Amazon region. New data from this area will be of crucial importance for resolving the biological scenarios that occurred at the Pleistocene/Holocene transition.

The question of why such a severe biological crisis did not happen at any of the previous glacial/interglacial transitions that characterized Quaternary history remains unresolved. Numerous attempts have been made to solve this crucial question (e.g. Graham and Lundelius, 1984; Graham et al., 1996). In contrast with those who maintain the 'overkill hypothesis' (for a review see Alroy, 2001) or simply blame global environmental changes (e.g. Van der Hammen, 1981; Van der Hammen and Absy, 1994), we believe the answer lies in the crucial role played by fundamental parameters: aridity, humidity and geographical factors, more than simply temperature changes. Alternations between cooling and warming were successively repeated, at differing intensities, during previous glacial/interglacial transitions. However, anomalous arid and humid conditions may have characterized the transition of the Pleistocene to the Holocene. The profound impact of these factors on the composition of the vegetation at the last glacial maximum/Holocene transition has been recently stressed by Behling and Hooghiemstra (1998, 1999), Van Der Hammen and Hooghiemstra (2000), and Baker et al. (2001) in study on Cordillera and Amazonian lake sediments.

The proposed scenario must be substantiated with data on the timing and pathways of megafauna migrations from the Cordillera to the refugia of the coastal region of Ecuador, which we hypothesize to have followed an increase in aridity. Findings in progress at Pueblo Nuevo (Machallilla), an Upper Pleistocene site, may provide evidence of crucial importance for the completion of the model. At Machallilla, the association of megatherids and mastodons suggests an ecological situation in transition toward the unbalanced ecosystem characterizing Santa Elena and other sites at the beginning of the Holocene.

We believe that the megafauna definitively abandoned the Cordillera areas around 12,000 yr BP, and migrated to coastal areas where they formed new faunal associations similar to the faunal assemblages at Machallilla (which was still open woodland at that point). As vegetation cover increased, the megafauna was concentrated in isolated areas where ecological conditions still were suitable, such as the Santa Elena Peninsula and Amazonian areas. However, the unusually high faunal concentrations and the change to dense vegetation cover at 8000–6000 yr BP, caused the final collapse and extinction of most elements of mammal megafauna.

We are conscious that such a conclusion is speculative, but our present research efforts are aimed at substantiating the model by means of bone collagen stable isotopic analyses sampled in well-dated faunas from different areas of the South American continent, which cover critical Quaternary climatic phases. In addition, isotopic and palynological analyses may reveal whether aridity/humidity values reached the hypothesized intensities at the Pleistocene/Holocene transition.

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